

The secret role of fascia in the martial arts

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Introduction

We live in two worlds, one on either side of our skin. The very survival of the Ninja or the hunting wild cat is dependent on their alertness and presence in both worlds. Body-Mindfulness is the term I use to refer to this embodied awareness and aliveness. It is both a state and a skill to be developed. Awareness of the body is the ground floor, the entry portal to mastery of the martial arts. Body-Mindfulness is a calm, yet alert, open state of present-time awareness of inner and outer body experiences, including sensory stimulations such as pressure, touch, stretch, temperature, pain, tingling, physical movement and position in space, visual, auditory, taste and olfactory impressions and even our inner organ and gut feelings. The care and curiosity required to refine this skill naturally develops a deeper mindfulness of the mind and heart, with all its inherent complexities.

The quest for ultimate power and awareness in martial arts is ancient. Two thousand years ago, Shaolin and Tai Ji masters training tendon power knew something intuitively that science has only recently validated. The masters recognized the vital importance of the appropriate conditioning and strengthening of the fascia and connective tissues to build and protect the body's Qi (life force). They operated from the assumption that the Qi of the universe and the Qi of the individual are one and that we can recharge our Qi directly from the universal source. They cultivated the

power of the physical body but also trained to circulate the Qi as light in the three Dan Tien, the head, heart and gut centers. They studied to support the body's own healing wisdom with touch transmission, herbal concoctions and acupuncture. Science now recognizes these centers as our three brains.



Figure 34.1

Shaolin Kung Fu training develops fascial strength.

SiFu Pierre Yves Roqueferre.

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What has not been appreciated about fascia until recently is how much we have to learn about its contractile properties, its elastic potential, and its important role as a global connector as well as a sensory awareness tissue. Research has confirmed that the body's extensive network of appropriately tensioned fascial sheets, tubes, membranes, wrappings, and fine and powerful cords, is not only the main connector for the functional framework but the fascia is "how" our body holds its form, senses, and achieves its function in stillness and in movement. It is the architectural tissue of the body and, in many ways, the home of our innerness, our self-sensing.

As most musculo-skeletal injuries involve inappropriate loading of the fascia and connective tissues, rather than the muscles, the fascia must be considered an important factor in training, peak performance and rehabilitation.

The new fascia researchers define fascia more broadly than traditionally. They recognize fascia as permeating the entire human body as one interconnected tensional network, including tendons, ligaments, joint and organ capsules, membranes, dense sheets and softer collagenous layers (Chapter 1). In the new fascia training there is an emphasis on developing elasticity, of acknowledging the stress-responsive nature and tensional integrity of fascial tissue, of conditioning and hydrating the fascia for appropriate stress-loading, as well as appreciating its proprioceptive and interoceptive qualities (Chapter 15).

This chapter explores current insights into the role of fascia in martial art training. Fascia's new sensory attributes elevates this previously unappreciated tissue to a much more significant role in our body-wide awareness and explains why fascial awareness is one of the secrets for success in martial arts mastery and for a successful human life.

The mindful heart and the right technology

The Qi generated and expressed in martial arts practice follows the deepest intention of our hearts and minds, and, according to Chia, our fascial planes. "The fasciae are extremely important in Iron Shirt Chi Kung; as the most pervasive tissue they are believed to be the means whereby Qi is distributed along acupuncture meridians" (Chia, 1988).

How do we explain an event where a light-framed woman somehow lifts a car to save her child? Her shock and desperate belief enabled



Figure 34.2

The Qi follows the intention of the heart and mind.

Harumi Tribble, dancer and choreographer.

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her to transcend her usual capacity. Her muscles were not strong enough to lift the car. How did she do it? Perhaps, like a kangaroo, which does not have enough muscle strength to jump very far but can spring great distances with tendon and fascia power (Chapter 8), her focused mind, coupled with tendon power, took her beyond what we would consider possible.

Lee Parore, former trainer for world heavyweight boxer, David Tua, sees awareness as primary: “Technology is the key to high performance but we are often looking at the wrong technology. The key isn’t machine training or muscle bulk. It is awareness and then self-awareness. It is a fire inside us that awakens the power in the Lower Dan Tien Energy Centre, below the navel. The lower abdominal musculature is double-layered because it’s the main switch box for absolute Qi power. This life force floods all the conductive fascial channels in the entire body for vitality and regeneration” (Parore, 2013).

Fascial awareness – Is our fascia feeling our muscles?

Schleip considers the connective tissues as the global connecting network, a listening system for the whole body: “Science now recognizes this richly innervated body-wide fascial web as the seat of our interoception, exteroception, kinaesthesia and proprioception, and hence the sensory base for our experience of embodiment” (Schleip, 2013). In fact, the fascia is six times more richly innervated than our muscles. So, when we say that our muscles are sore, it may be that we are feeling our fascia or that our fascia is feeling our muscles (Chapter 15).

Cirque du Soleil acrobat, Marie Laure, related that her refined body awareness was even more important than her core strength and endurance,

in her performances with her partner. One might assume that her impressive muscular power would translate into rigidity in her fascial and muscle tissue. On the contrary, as a therapist, I experienced her myofascial tissue to be surprisingly supple and her physique hardly bulked. McGill says, “The hallmark of a great athlete is the ability to contract and relax a muscle quickly and to train the rate of relaxation” (McGill, 2004).

“Measuring top mixed martial arts competitors, the ones who hit the hardest were not the ones with the biggest muscles. When a muscle contracts it creates force and stiffness- stiffness slows motion and velocity. The most powerful athletes use skilled strategic muscle pulses to create the highest speed and most effective force. These pulses are potentiated with strategic storage and recovery of elastic forces into the fascial system. A tuned fascial system is essential for high performance. The fighters with big muscles ‘pushed’ their punches and had less impact” (McGill, 2018).

Fascial awareness enables the martial artist to self-tune the muscles and fascia and maintain high alertness in their internal and external worlds. This is a practice of feeling with the whole body, sensing nuances in the balance of the other’s body and then manipulating or deflecting the other’s expressed energy to their own advantage.

Huang Sheng Shyan was famous for his powerful issuing of the Tai Ji-Jin elastic force. His training method involves precise attention to the internal changes of muscles, tendons and fascia. Patrick Kelly, Tai Ji teacher and author, speaks about this awareness, where internal relaxation means every muscle in the body elongates and actively stretches under the pressure, rather than contracts and shortens (or holds unchanged) in

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a tense resistance. “We can say that the basis of the secret is: Tai Ji-Jin is motivated by the Yi (mind intention) energised by the Qi, issued from the root and transmitted through the body in a wave of stretching muscles” (Kelly, 2007). Intense physical training, without self-awareness, will not produce the best results. There are many stories of masters stopping an opponent with their mind intention alone. This is the inner art. If we do not develop our inner and outer sensing, we will not achieve awareness of the space around us, of our opponent, or of the sword we hold. Quite simply, our fascia is a vital sensory organ in movement and martial training.

Building fascial resilience and fascial armoring

Iron Shirt is an ancient Kung Fu method that profoundly takes advantage of the stress-responsive nature of the bones and fascia. Wolff’s law states that bone, in a healthy person or animal, will adapt to the loads under which it is placed. If loading increases, the bone remodels itself over time to become stronger. Iron Shirt strengthens muscles, tendons, fascial sheaths and bones, by subjecting them directly and gradually to increasing stress. The important detail here is “gradually”. Internal energy visualization, combined with breath and body conditioning, are used to cultivate the Qi. “The Qi that is generated is then stored in the fascial layers where it works like a cushion to protect the organs” (Chia, 1988).

This concept of Qi between tissue layers helps to explain an experience I had with a Singaporean Tai Ji master in 1981. I was waiting at a table to ask if I could study with him. He walked out, smiled, reached across, pinching the skin of my forearm between his thumb and forefinger, rolling it back and forth in different places for about 10 seconds. “Yes,” he said, “I can see you have

been practicing Tai Ji for a while. Good practice creates elasticity between the tissue layers.” Lau Chin-Wah was a senior student of Master Huang. I studied with him in Kuching, Malaysia. Our practice centered on refining the form, Pushing Hands and White Crane Quick Fist. Sometimes Chin-Wah would throw his arm, as if it was an enormous wet rag or rope, against a stone wall, apparently not hurting himself at all. He said the secret was to totally relax the muscles and fascia of the arm and to condition and harden the bones to become like steel, by training with a partner, hitting bony forearms and shins against each other. This is extremely painful initially but becomes easier and may transform even the marrow and substance of the bones and soft tissues (Chia, 1988). Chin-Wah spoke of the cultivated Qi inside him, gathering, without his will, to meet the force of an attack. He said that when he was centered in his structure, his body and energy field, his fascia, bones and soft tissues became like impenetrable armor, enabling him to have a heavy roof tile broken over his head as if it were nothing. Our fascial system is capable of great adaptation to precise and progressive loading.

Fascia power training keys

Training for speed depends on our capacity to relax

In boxing drills, fighters need to become experts at letting go of all muscle tension. A system that is genuinely relaxed does not need to overcome tension to ignite itself for immediate response. “You don’t want muscle bulk for true speed and strength in boxing but resilient fascia like Spiderman” (Parore, 2013). In Tai Ji Pushing Hands, the players must be willing to be pushed over and invest in loss, to learn to use elastic fascia force instead of relying on muscular

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strength to combat force. Master Huang was simultaneously totally relaxed yet totally stable. To push him was to feel drawn into empty space and then be sent flying.

Bruce Lee, the famous Kung Fu master, was not a big man. At only 1.7 m and 68 kg, he often defeated opponents who were both larger and much heavier. One day, he demonstrated his one-inch punch to a skeptical, burly man, who was astonished as he flew back 5 m into the swimming pool. McGill's perspective on this was that Bruce Lee did not hit with only the mass of his fist, but rather that his entire body hardened into an effective mass, so the fascia allowed him to hit his opponent with 68 kg (McGill, 2018).

When Lee himself was asked how he had done it, he said, "To generate great power you must first totally relax and gather your strength, and then concentrate your mind and all your strength on hitting your target" (Hyams, 1982).

The body's mind modifies your peripersonal space

Neuroscientists call the space around the body peripersonal space (Rizzolatti et al., 1997). Recently, brain mapping techniques have confirmed what my first Tai Ji Master said to me: "You must extend your feeling sense out around your body. When you move the sword through space, feel the length of it and the tip, as if it was part of your body" (Mak Ying Po, 1976). Similarly, in *The Body Has a Mind of Its Own* the Blakeslees, say, "Your self does not end where your flesh ends, but suffuses and blends with the world, including other beings. Thus when you ride a horse with confidence and skill, your body maps and the horse's body maps are blended in shared space" (Blakeslee and Blakeslee, 2007). The kick-boxer and the Kung Fu master both rely on the sensing mechanism of their body-wide fascial

network to feel both their own self and the other. This enables them to respond without hesitation to the slightest change in their peripersonal space. As Master Huang said, "If you are thinking, it's too late" (Huang Sheng Shyan, 1980). The most profound training takes us beyond thinking to a new automatic responsiveness.

Tensegrity strength: stabilizing the myofascial framework

Those studying the fascial matrix have compared the body to a bio-tensegrity structure, a living structure of tensional integrity. Unlike a pile of bricks, our bones do not touch each other but are spaced by cartilage or soft tissues and the skeletal relationships are maintained by the tension and span of the global architecture of the myofascial system, in some ways like a tent. The concept of spreading the stress and tension throughout the whole body may shed light on how barefoot runners may endure less shin splints than runners in protective shoes (Warburton, 2001). Barefoot runners strike the ground more gently with more forefoot than runners in conventional shoes. This reduces the stress transmitted to the shin-bones and joints and spreads it throughout the entire fascial and skeletal framework, where it is stored as elastic energy (Chapter 32). Balance exercises, such as slack line, wobble board, Swiss ball, and rock climbing, challenge and train our fascia and internal strength to develop spontaneous tensegral adaptability.

I observed Cirque du Soleil acrobats maintaining their core stabilization through "animal-like" stretches and play, consistent practice of their art, and specific exercises to strengthen the flexors, extensors, lateral torso and hip muscles for balance of myofascial tone in all three functional planes. They rarely seemed to train with machines. This core stabilization and core

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stiffening is essential for hyper-mobile acrobats to avoid injury. The capacity for a resilient yet mobile whole body stabilization is paramount for martial artists.

The no inch punch: preparatory counter movement at its most subtle

In boxing and Wing Chun punching training, much like the golf swing, we see the value in spring-loading and pre-tensioning to unleash the explosive punch. In fighting, it is vital that our intention is not telegraphed to our opponent. Perhaps the alert poise of the kick-boxer could be seen as a sustained pre-stretching of the fascial network, where the muscles prevent the release of recoil-power until the precise moment for the perfect delivery (Chapter 8).

Alan Roberts, Aikido teacher said, “In Aikido, Cheng Hsin and Jiu Jitsu, much of the purpose of the internal training is to develop the ability to strike with immediacy and power. This is even more of a concern in the sword arts, where efficiency, speed, accuracy and the unexpected are highly prized” (Roberts, 2013).

Huang was known throughout the Chinese martial arts world for his capacity to effortlessly throw opponents many meters. Not only was he not extending his hands and arms to do so, but, paradoxically, his hands almost appeared to be withdrawing as the person flew back, as if from an electric shock. In Tai Ji, the deep stance, focused breath and cultivation of Qi in the lower Dan Tien, as the weight is shifted, is a systematic spring-loading, ready to explode the issuing force.

Peter Ralston, author of *Cheng Hsin: The Principles of Effortless Power* (1989), when asked about Bruce Lee’s one-inch punch, placed his hand on the questioner’s chest and said, “You don’t need an inch,” then knocked him 6 m across the room, with little visible movement.

Actively stretching the fascia

Stretching is often seen as an important part of training. Yet how shall we best stretch? The classic holding of long passive stretches is not something we observe much in powerful primates or other animals. Naturally and spontaneously, animals roll, actively stretch against, scratch, push and rub their soft tissues and joints against the ground or trees, to neutralize built-up stress, and possibly to nourish and rehydrate their myofascial systems (Bertolucci, 2011). Interestingly, no other large primates in the wild appear to suffer as much from osteoarthritis or rheumatoid arthritis as much as modern-day humans do.

Our body is designed for active loading, walking, running, carrying, digging, climbing, etc. “We are hard-wired to move, for survival, pleasure, creative self-expression and optimal function. The body has an inherent vocabulary of movement patterns that develop naturally and concurrently with brain development and practiced coordinated movements. I refer to these as primary movements and postures. In cultures where people squat, sit cross-legged on the floor as a matter of course and walk barefoot at least sometimes, they cultivate their internal strength and fascial flexibility, which lasts into old age. On the other hand, many Westerners, even in their teens and certainly as they get older, struggle to squat flat-footed and sit upright on the floor. This loss of primary postures, movements and healthy fascial functioning is more and more evident in our chair-based society” (Petersen, 2009).

Our movement repertoire is locked into the myofascial mechanisms of our breathing patterns. Each breath is a physical and energetic impulse into the entire fascial network (Chapter 8). In martial arts, abdominal, reverse abdominal and conscious breathing is an integral part of a practice. The master realizes we are “breathed”

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by the universe and entering the universe of the breath is a prominent path within all meditation methods. Training of a powerful range of motion of the arm, spinal and leg myofascial chains in kicks, punches, blocks and evasive movements, amplifies the fascial elasticity and tone in both the core, the powerhouse of our internal strength, the extension of the three chains, and in the engine of our entire breathing mechanism.

Many trainers suggest not stretching immediately before demanding exercise but to warm



Figure 34.3

Demonstrates energizing the Qi in long anterior myofascial chains

Master Li Jun Feng.

up and mobilize the joints and tissues instead. Experience suggests that fast, dynamic stretching, which occurs in many kicks and punches, is beneficial for the fascia when performed correctly: soft tissues should be warm and abrupt movements avoided. Rhythmic controlled bouncing at the end range may also be effective, but for the elastic recoil, not the muscles (Chapter 8).

“Fascia research highlights the fusion of passive and active tissue. Tuning the interaction of these tissues is a higher concept than simply stretching. This enables optimal strength, speed, and power while reducing injurious stress concentrations. For example, when jumping: if the hamstrings are overstretched, only the active component of the muscle can create force. But great jumpers often have tighter hamstrings where they can time the muscle recoil with the elastic recoil of fascia and connective tissue, creating a higher resultant force. Thus more stretching is not the answer. Enough mobility is needed for the task but no more. The muscle creates force and so does tuned fascia. Consequently we have a better result” (McGill, 2013).

Train the fascia through Kata or Forms for total warrior fighting strength

“To prepare for mixed martial arts or real fight situations, it is important to realize, that functional strength can only be developed through exercises that not only work major muscle groups but also improve the condition and flexibility of the fascial planes,” says 8th degree black belt Grand Master, Lance Strong. “Kata or Forms training has a huge effect on developing fascial strength and your ability to apply that strength in many different directions, while still maintaining your body’s center and balance. Virtually no form of exercise, other than kata, tai ji or yoga, and some cross-training exercises, develops this ability” (Strong, 2013).

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Conditioning the fascial body

It takes time to build fascial resilience: “It takes two to three years to build an aikido body that is elastic and strong enough for the training practice. Most injuries occur in the tendons and ligaments and are caused by over-training too early or trying to practice techniques more roughly than is appropriate” (Roberts, 2013).

The Shaolin monks knew that, for positive results and to avoid injury, they could only do the intense bone conditioning every second or third day. We now know that collagen has a slow renewal cycle with a half-life of approximately 1 year. Fascial tissues, particularly tendons, have a very slow renewal cycle. We may have little to show initially but then more after 6 to 12 months (Chapter 1). In fascia training, appropriate loading, and in particular eccentric loading, may be more important than repetition. With free weights or kettle bells (Chapter 22), medium weights are used to train the fascia, and heavier weights are used for the muscles.

To achieve the best results, fascia training is limited to two or three times per week. On the other hand, daily exercise has great value for our brains, muscles and cardiovascular systems. So we need to be careful with short intense boot camp trainings, as this often propagates compartment syndromes and fascial inflammation. We evolved to be physically active endurance athletes. So our new recognition of the potential of the fascia in movement and exercise should not distract us from appreciating the importance and absolute necessity of training our muscles. The fascia and the muscles are married, totally connected. To embody our physical potential, we must train throughout our life the whole myofascial network and nurture the whole body-being.

“If tolerated by the joints, training should include a component that demands 100% neural

drive to the muscles. This is usually accomplished by training at speed. The actual static load need not be that great. Good form in all exercise promotes sparing of the joints but ensures optimal tensioning all along the musculoskeletal linkage” (McGill, 2013). McGill gives the example of a slow grinding bench press versus standing and falling into a push-up position, where the hands are on a low box, then immediately exploding back up to the standing position. The box height is adjusted so that this is barely possible. The neural drive is exceptional, as is the tuning of the fascia, passive tissue and active muscle tissue.

Finding the gift in an injury

In *Gift of Injury*, McGill documents a champion power-lifter, brought to his deepest despair with a damaged spine and sacrum, who applied the principles of rebuilding and repairing his skeletal and myofacial system to regain world-class performance. Many recognize the importance of assessment to guide appropriate training. When the system is weakened through illness or injury, a new motivation and a new training regime is required.

Finding the optimal dose of stimulation and stress (magnitude, duration, repetition and rest) is critical. Too little stimulation creates no effect and too much stimulation causes stress that damages cells. “Great clinicians hone the art of training to create the best adaptation for athletes at all levels” (McGill, 2018). “Mindfulness practice has a strong correlation with pain reduction making it an important skill in rehabilitation and trauma recovery” (Schleip, 2018).

Healthy fascia for a healthy body

The fascial web, rather than the muscles alone, provides a framework for storage and release of kinetic energy (Chapter 10). It governs the spring-loaded joint mechanisms for the leg, arm, spinal

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chains, the internal organs and the entire system. Without good nutrition and hydration, the fascia will let you down.

“The foundation for top performance has to be health. So before serious training with an athlete, I test the liver, heart, blood, and immune response and get the nutrition right so the Qi can move through the body. The fascia is a piezoelectric conductive collagen network, that when coupled with water, is the body’s battery for energy. The highest concentration of mitochondrial DNA is housed in the brain, the heart, and the immune system within our gut – the three classic Dan Tien energy centres. The mitochondrial DNA are the power plants of our cells with the nuclear DNA being the structural blueprint” (Parore, 2013).

The body’s acid/alkaline balance, hormonal influences, lymph and blood flow all have a strong effect on the fascia and this affects our fitness. We rehydrate fascia with an inclusion of appropriate resting times for tissue and viscoelastic recovery. Vitamin C is important in collagen formation. Scurvy, a connective tissue disease, occurs when there is a deficiency. An active life and light nutrition support the fascia and muscles better than a sedentary lifestyle. On a heavy red meat diet with saturated fats and refined sugar products, we may cultivate stress hormones and chronic inflammation.

Fascial restrictions and scarring can affect and inhibit a martial arts practice (Chapter 2). For self-care maintenance and injury healing, foam rollers, massage balls and self-massage can be helpful to release restrictions that can compromise our function. Some therapists use stone massage tools or powerful herbal liniments to reduce fascial scarring. New interventions such as injecting medical ozone, vitamin cocktails and prolotherapy assist in fascial strengthening

and recovery. Adhesions can also be released with integrative myofascial approaches such as osteopathy, acupuncture, integral aquatic therapy and structural integration.

Body-Mindfulness – training for the art of life

The path of the ultimate warrior is a path of awareness, service, love and protection for our community. It demands one to access the deepest qualities of the warrior: focus, energy, perseverance and dedication to a cause bigger than oneself. This heart, this spirit, is the essence of the internal power of the martial arts. It is essential for reaching peak performance, recovering from an injury, or building resilience to deal with the inherent challenges of life. The Buddha said, “Mindfulness is the sole path to freedom” (Goldstein, 1976). He outlined four pillars for the practice of mindfulness. The first two focus on mindfulness of the body, with the fascia now highlighted as the means of our proprioception, kinesthesia, exteroception and interoception. The second two pillars are mindfulness of the mind: the functioning mind we identify as the “me” and mindfulness of the path beyond “me” to the true Tai Ji – unity consciousness.

Contemporary Western culture and the media tend to draw us away from deeper body awareness. A committed martial arts practice is a daily returning to our self. True masters are rare beings who have achieved the ultimate physical expression and the deepest Body-Mindfulness, with the consequent fascial awareness and peace of mind. This is built step-by-step during a lifetime of training as they integrate their art into every dimension of their life and well-being. As Master Huang said, “Eat, sleep and practice Tai Ji” (Huang Sheng Shyan, 1980).

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Be assured, Body-Mindfulness need not be a serious task. To expand your myofascial repertoire is to become elastically playful. It involves bringing a new creative attitude, not only to your martial arts practice but also to simple everyday activities like standing on one leg to brush your teeth, getting comfortable with sitting on the floor, studying stretching with a local cat, and regular practice of the flat-footed squat. It is never too late to begin training. My student Stella started Tai Ji aged 85. She struggled to climb the stairs to the practice room and to accomplish basic movements. At 88, her practice had lightness and flow, and her balance

had improved impressively. By 93 she carried her success in every step, reminding us of the adaptability of the neuro-myofascial body, even in old age.

In conclusion, to bring awareness to our fascia is to become more present and more Body-Mindful. This foundation of attention naturally leads us to more consciousness of our mental, emotional and spiritual selves. Cultivating Body-Mindfulness in a martial arts practice brings a fresh aliveness, an enthusiastic resilient energy, and a generous self-care attitude: tools we all need for a long and healthy life.

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